1. Define Knowledge Base.

A Knowledge Base (KB) is a collection of sentences that represent facts about the world in a knowledge representation language. It is a central component of a knowledge-based agent.

Example:

- Human(Sita) — Sita is a human.
- $\forall x$ Human(x) $\Rightarrow$ Mortal(x) — All humans are mortal.

Using these facts and rules, the agent can infer that Sita is mortal.

2. Describe Knowledge Representation Language.

A Knowledge Representation Language (KRL) is a formal language used to represent facts, objects, relationships, and rules about the world in a knowledge base.

It consists of:

- Syntax: Specifies the valid sentences in the language.
- Semantics: Defines the meaning and truth of sentences in a model.

Examples include propositional logic and First-Order Logic (FOL).

3. Give the differences between Forward and Backward Chaining.

Forward Chaining	**Backward Chaining**
Starts with known facts.	Starts with a goal or query.
Applies rules to derive new facts.	Searches for facts and rules that prove the goal.
Data-driven approach.	Goal-driven approach.
Useful when new information is received.	Useful for answering specific queries.

Example: Forward chaining derives Mortal(Sita) from known facts, whereas backward chaining starts with the goal Mortal(Sita) and searches for supporting facts and rules.

4. Define Inference Rule.

An inference rule is a logical rule used to derive a new sentence or conclusion from existing facts or sentences in a knowledge base.

Example — Modus Ponens:

- $P \Rightarrow Q$
- P
- Therefore, Q.

- If all humans are mortal and Ravi is human, we can infer that Ravi is mortal.

5. Apply Inference Rules for Quantifiers.

First-Order Logic uses inference rules to derive conclusions from universally and existentially quantified sentences.

- Universal Instantiation (UI): Replaces a universally quantified variable with a specific ground term.
- Existential Instantiation (EI): Replaces an existentially quantified variable with a new constant.

Example:

Given $\forall x \; Human(x) \Rightarrow Mortal(x)$, applying UI with $x = Ravi$ gives:

$$Human(Ravi) \Rightarrow Mortal(Ravi)$$

If `Human(Ravi)` is true, we infer `Mortal(Ravi)`.

6. Define Universal Instantiation.

Universal Instantiation (UI) is an inference rule that allows us to substitute a ground term for a universally quantified variable.

General form:

$\forall x \, P(x) \Rightarrow P(a)$ Here, a is a ground term.

Example:

$$\forall x \; Human(x) \Rightarrow Mortal(x)$$

Substituting $x = Ravi$, we obtain:

$$Human(Ravi) \Rightarrow Mortal(Ravi)$$

7. How is Skolemization applied in Knowledge Representation?

Skolemization is the process of eliminating existential quantifiers by replacing existentially quantified variables with new Skolem constants or Skolem functions.

- A new constant is used when the existential variable does not depend on universally quantified variables.
- A Skolem function is used when it depends on universally quantified variables.

Example:

$$\exists x \; Human(x)$$

After Skolemization:

$$Human(A)$$

Here, A is a new Skolem constant. Skolemization is used while converting First-Order Logic sentences into Conjunctive Normal Form (CNF) for resolution.

8. Define Semantics.

Semantics defines the meaning of logical sentences and specifies whether a sentence is true or false in a particular model.

In propositional logic, semantics assigns truth values to proposition symbols and determines the truth values of complex sentences using logical connectives.

Example: If $P = True$ and $Q = False$, then $P \wedge Q = False$.

9. Give the differences between Propositional and First-Order Inference.

Propositional Inference	**First-Order Inference**
Works with complete propositions.	Works with objects, predicates, variables, and relations.
Does not use quantifiers.	Uses universal (\(\forall\)) and existential (\(\exists\)) quantifiers.
Uses propositional inference rules.	Uses rules such as UI, EI, and generalized Modus Ponens.
Cannot directly express general rules about individual objects.	Can represent general rules about objects and their relationships.

Example: FOL can represent "All humans are mortal" as $\forall x \; Human(x) \Rightarrow Mortal(x)$.

10. Define Resolution.

Resolution is an inference rule used to derive conclusions by eliminating complementary literals from two clauses.

Example:

$P \vee Q$

$\neg P \vee R$

Resolving on P and $\neg P$ gives:

$Q \vee R$

Resolution is widely used in automated theorem proving and can prove a statement by deriving a contradiction from its negation.

11. Describe the Role of Lifting in Knowledge Representation.

Lifting extends propositional inference methods to First-Order Logic by allowing inference directly on sentences containing variables rather than generating every possible ground instance.

It uses unification to find substitutions that make logical expressions match.

Example:

Example:

- `Human(x) ⇒ Mortal(x)`
- `Human(Ravi)`

Using the substitution $x = Ravi$, we infer `Mortal(Ravi)`.

Lifting reduces unnecessary instantiations and makes first-order inference more efficient.

12. Define Equality and Tuple.

Equality: In First-Order Logic, equality (=) asserts that two terms refer to the same object.

Example: Father(John) = Henry means John's father is Henry.

Tuple: A tuple is an ordered collection of objects used to represent a relationship between them.

Example: $\langle Richard, John \rangle$ represents an ordered pair in the Brother relation if Richard is John's brother.

13. Define Ground Sentences.

A ground sentence is a logical sentence that contains no variables. It uses constants and ground terms instead of variables.

Examples:

- Human(Ravi) — Ground sentence.
- Mortal(John) — Ground sentence.
- Human(x) — Not a ground sentence because it contains the variable \(x\).

Ground sentences are useful in propositionalization and logical inference.

14. Define Lifting Lemma.

The Lifting Lemma states that if a propositional resolution step can be performed on ground instances of two First-Order Logic clauses, then a corresponding first-order resolution step can be performed on the original clauses using an appropriate unifier.

It establishes the connection between ground-level resolution and first-order resolution and is used to prove the completeness of first-order resolution.

15. Define Entailment.

Entailment means that a sentence logically follows from a knowledge base.

It is represented as:

$$KB \models \alpha$$

This means that α is true in every model in which the knowledge base KB is true.

Example: If KB contains `Human(Ravi)` and `Human(x) ⇒ Mortal(x)`, then:

$$KB \models Mortal(Ravi)$$

16. Define Model.

A model is a mathematical representation of a possible world that specifies whether logical sentences are true or false.

In propositional logic, a model assigns truth values to proposition symbols. In First-Order Logic, a model consists of a domain of objects and an interpretation of constants, predicates, and functions.

Example: A model may assign Human(Ravi) = True and Mortal(Ravi) = True.

17. Define Completeness.

An inference method is complete if it can derive every conclusion that logically follows from the knowledge base.

In other words, whenever:

$$KB \models \alpha$$

a complete inference procedure can derive α from KB.

Completeness ensures that logically valid conclusions are not missed by the inference method.

18. Define Grounding.

Grounding is the process of connecting the logical knowledge maintained by an agent to the real-world environment.

An agent uses sensors to obtain observations and adds corresponding facts to its knowledge base.

Example: In the Wumpus World, when an agent senses a stench, it adds the corresponding percept to its knowledge base and uses it to reason about the possible location of the Wumpus.

19. Define Sound or Truth-Preserving Inference.

An inference method is sound or truth-preserving if every conclusion it derives is logically entailed by the knowledge base.

Formally, if the inference procedure derives α from KB, then:

$$KB \vdash \alpha \Rightarrow KB \models \alpha$$

Example: From $P \Rightarrow Q$ and P deriving Q using Modus Ponens is sound.

20. Explain Operations in a Knowledge Base.

A knowledge-based agent performs three main operations:

1. TELL: Adds information or percepts to the knowledge base.
2. ASK: Queries the knowledge base to determine what action should be taken or what conclusion follows.
3. PERFORM: Executes the selected action in the environment.

These operations enable the agent to update its knowledge, reason about the environment, and act intelligently.

21. Define Propositional Logic.

Propositional Logic is a formal system in which knowledge is represented using propositions that can be either true or false.

Complex propositions are formed using logical connectives such as:

- Negation ($\neg$)
- Conjunction ($\wedge$)
- Disjunction ($\vee$)
- Implication ($\Rightarrow$)
- Biconditional ($\Leftrightarrow$)

Example: Let P denote "It is raining" and Q denote "The road is wet." The statement "If it is raining, the road is wet" is represented as $P \Rightarrow Q$.

22. Differentiate FOL and Propositional Logic.

Propositional Logic	**First-Order Logic (FOL)**
Represents facts as complete propositions.	Represents objects, properties, and relationships.
Uses proposition symbols such as P and Q.	Uses predicates, variables, constants, and functions.
Does not support quantifiers.	Supports universal and existential quantifiers.
Has limited expressive power.	Has greater expressive power.

- Propositional Logic: P — Ravi is a student.
- FOL: $\forall x \; Student(x) \Rightarrow Learner(x)$ — All students are learners.

23. State One Limitation of Propositional Logic.

One major limitation of propositional logic is its limited expressive power. It cannot directly represent variables, quantifiers, or general relationships between individual objects.

Example: To represent "All humans are mortal," propositional logic would require separate propositions for individual humans, whereas FOL expresses it compactly as:

$$\forall x \; Human(x) \Rightarrow Mortal(x)$$

Thus, FOL is more suitable for representing general knowledge about objects and their relationships.

24. Give an Example of an Inference Rule.

Modus Ponens is a commonly used inference rule that derives a conclusion from an implication and its premise.

Rule:

$$\frac{P \Rightarrow Q, \quad P}{Q}$$

Example:

1. If a person is human, then the person is mortal: $Human(Ravi) \Rightarrow Mortal(Ravi).$
2. Ravi is human $Human(Ravi)$
3. Therefore, Ravi is mortal: $Mortal(Ravi)$

   Hence, Modus Ponens derives a valid conclusion from the given facts and rule.